HW 7

problem 3:

Part c) error in stellar radius $\sigma_{R_*} = 0.01 R_\odot$, error in fractional depth $\sigma_\delta = 0.002$

Find error in planetary radius, σ_{R_p}

$$\sigma_{R_p} = \sqrt{\left(\frac{\partial F}{\partial R_*}\sigma_{R_*}\right)^2 + \left(\frac{\partial F}{\partial \delta}\sigma_\delta\right)^2} \qquad F = R_p = R_* \sqrt{\delta}$$

$$R_* = 1.14 R_\odot, \quad \delta = 0.0097865$$

$$\sigma_{R_p} = \sqrt{\left(\sqrt{\delta} \cdot 0.01\right)^2 + \left(\frac{1.14}{2\sqrt{\delta}} \cdot 0.002\right)^2} = \boxed{0.0115 R_\odot}$$